Ordinary Polynomial Multiplication over Fixed Finite Fields in $O(n \log n)$ Time

A baseline companion and fixed-basis reduction

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Abstract

For every individually fixed finite field $\mathbb{F}_q$, we derive ordinary dense polynomial multiplication in $O_q(n \log(n+2))$ deterministic bit time on a fixed finite-alphabet Turing machine with a fixed finite number of sequential tapes. The characteristic-two branch uses the accompanying Pascal-transform binary multiplier; the odd-prime branch weakens the accompanying exponent 9999/10000. A fixed-basis expansion reduces a degree-e extension to e^2 prime-field products and $O_q(n)$ encoding and recombination work, including ordinary coefficient input/output. This note records a baseline consequence of shared proof sources. Complete Lean formalization is unfinished.

AI assistance and review status. This work was developed with substantial generative AI assistance in mathematical exploration, proof development, writing, coding and source review. OpenAI Codex was used for release preparation and source inspection; the platform reference [9] identifies the tool, not a mathematical premise. A complete historical model/version log is not available. Full Lean formalization is unfinished and planned for a later update, with no completion date promised. This preprint is open for mathematical review; independent human proof review is not claimed.

1 Scope, prior work and the role of this note

The question is the bit complexity of multiplying ordinary dense univariate polynomials over an individually fixed finite field. Harvey, van der Hoeven and Lecerf established a Fürer-type bound with the factor $8^{\log^* n}$ for each fixed prime field [1]. Harvey and van der Hoeven then obtained the unconditional fixed-prime bound

$$O_p\Big(n \log n \, 4^{\log^* n}\Big)$$

in the deterministic multitape Turing model [2, Theorem 1.1]. For a fixed extension degree the same asymptotic form follows by coordinate reduction. These are important predecessors for the fixed-field comparison in this note.

Their later finite-field theorem gives $O(n \log q \log(n \log q))$ uniformly in q, under the hypothesis that a Linnik constant $L < 1+2^{-1162}$ exists [3, Theorem 1.2]. Thus its fixed-q specialization is a conditional $O_q(n \log n)$ theorem. Its quantifiers and number-theoretic hypothesis should remain explicit. Harvey's October 2026 discussion separately identifies fixed-field polynomial multiplication as a problem not directly settled by the new integer multiplication claim, and records the unconditional cyclotomic benchmark [4, Section 4, p. 4]. That discussion supplies context, not verification or endorsement of this note.

This history explains the role of the $n \log n$ benchmark: the unconditional fixed-field algorithms reached near-linear time, while the sharper uniform theorem retained its explicit hypothesis. The accompanying new constructions claim to close the remaining fixed-field gap. This note makes their shared baseline consequence and the elementary extension-field reduction easy to identify.

The present baseline combines two arguments in the accompanying preprints. For characteristic two it uses the Pascal-transform route of [5]. For odd characteristic it uses only the odd-field proposition of [6]. It does not use that paper's stronger combined all-field theorem or the separate controlled-CRT binary multiplier. The fixed-basis reduction below makes the resulting field quantifiers and ordinary input/output boundary explicit. This is a companion consequence of shared proof sources, not a separate independent discovery or a practical speed claim.

2 The fixed-field statement

Let $q = p^e$ be fixed, with p prime and $e \geq 1$. Choose a fixed $\mathbb{F}_p$-basis $b_0, \ldots, b_{e-1}$ of $\mathbb{F}_q$. Represent a field element by its e prime-field coordinates, each in a fixed binary encoding. The field, basis and structure constants are fixed before the input length varies; they are not growing inputs or advice depending on n.

An input is the two ascending coefficient lists of polynomials $A, B \in \mathbb{F}_q[X]$ of degree less than n, with $n \geq 1$, including leading zero coefficients. The required output is the ascending list of all $2n-1$ coefficients of AB, also retaining leading zero coefficients. Input and output have $\Theta_q(n)$ bits. Time includes all encoding, setup, movement, component calls, decoding and output.

Theorem 1 (Ordinary fixed-field baseline)**.** *For every individually fixed finite field $\mathbb{F}_q$, the proof routes in [5, 6] give a deterministic algorithm for this ordinary product in*

$$O_q(n \log(n+2))$$

bit steps. The machine has a fixed finite alphabet and a fixed finite number of sequential one-dimensional tapes. Its finite control, tape count and implied constants may depend on q, but not on n.

3 Prime-field inputs and inherited dependencies

Write $M_p(n)$ for the cost of an ordinary prime-field multiplier with the input/output convention just stated.

For $p = 2$, the completed reduction and binary tensor construction in [5] state $M_2(n) = O(n \log(n+2))$. That route includes its own evaluation, pointwise product and residue decoder. It retains the arbitrary-width equal-chunk tape-interchange result from OpenAI's preprint [7, Lemma 12, Section 4]. This note inherits that cited theorem dependency; it does not infer a binary polynomial multiplier simply from an integer-multiplication bound.

For an odd prime p, Proposition 1 of [6], specialized to extension degree one, states

$$M_p(n) = O_p\Big(n(\log(n+2))^{9999/10000}\Big).$$

Its bundled odd-field proof uses the unconditional cyclotomic host of [2], the accompanying native constructions and a named stage-two interchange theorem [8]. The proof sources and exact movement adapter are identified in that companion's `proofs/PROOF_SOURCES.md`. This odd proposition is proved separately from its characteristic-two combination.

Since $\log_2(n+2) > 1$ for $n \geq 1$ and $9999/10000 < 1$, weakening this bound gives $M_p(n) = O_p(n \log(n+2))$. In particular, the odd part of this baseline does not introduce a new Linnik-type hypothesis. The present note relies on the written cited arguments; it does not independently certify their correctness or formalize their full machine constructions.

4 Fixed-basis reduction and tape costs

Lemma 1 (Constant-degree coordinate reduction)**.** *For a fixed extension $\mathbb{F}_q/\mathbb{F}_p$ of degree e, an ordinary prime-field multiplier of cost $M_p(n)$ yields an ordinary $\mathbb{F}_q$ multiplier of cost*

$$O_q\Big(e^2 M_p(n) + n\Big).$$

Here e is fixed; displaying it records the number of component products, not a uniform growing-extension complexity claim.

Proof. Expand the coefficient lists in the fixed basis to write

$$A = \sum_{i=0}^{e-1} b_i A_i, \qquad B = \sum_{j=0}^{e-1} b_j B_j, \qquad A_i, B_j \in \mathbb{F}_p[X], \quad \deg A_i, \deg B_j < n.$$

Each A_i and B_j is an ordinary coefficient list in ascending degree order. Fixed-width coefficient decoding and a fixed number of sequential scans write these $2e$ lists in $O_q(n)$ steps. An initial length scan and all component-call descriptors have the same bound. Any other fixed field-element encoding can be converted by fixed finite tables at this cost.

Compute the e^2 ordinary products

$$C_{ij} = A_i B_j \quad (0 \leq i, j < e)$$

with the prime-field multiplier. Each result has $2n-1$ ordinary coefficients. Calls and input copies are sequential, and the fixed number of component machines can be incorporated into one finite control with a fixed finite set of work tapes. Their total cost, including their stated setup and cleanup, is $e^2 M_p(n) + O_q(n)$.

The fixed structure constants $\mu_{ijr} \in \mathbb{F}_p$ satisfy

$$b_i b_j = \sum_{r=0}^{e-1} \mu_{ijr} b_r.$$

Consequently, if $D_r = \sum_{i,j} \mu_{ijr} C_{ij}$, then $AB = \sum_r b_r D_r$. Recombination uses a fixed number of scans over the component lists: the k-th output is formed only from their k-th coefficients. All operations on $\mathbb{F}_p$ and all fixed scalar multiplications have constant bit cost depending on q. At most $O(e^3 n) = O_q(n)$ such work is required. A fixed number of tapes retains the component streams, so these scans assume no unit-cost random access or free global data permutation.

Finally encode $\sum_r b_r (D_r)_k$ and write it for $k = 0, \ldots, 2n-2$. This is precisely the ordinary product list; no cyclic wrap, transformed output or missing high coefficients remain. Output writing and disposal of temporary streams cost $O_q(n)$, proving the stated bound. □

Proof of Theorem 1. Section 3 supplies $M_p(n) = O_p(n \log(n+2))$ for each fixed prime p, with the cited dependency chain appropriate to its characteristic. Apply Lemma 1. Since p, e and all field data are fixed, e^2 is constant and the linear codec/recombination term fits the stated bound. The finitely many small lengths can be handled by ordinary finite-table arithmetic on the same machine. □

5 Verification and relation to stronger claims

This note isolates the general $n \log n$ baseline and its fixed-basis proof. The stronger general companion [6] states a strict logarithmic saving by a different combination of its odd-field proposition with the principal binary paper. The baseline proved here instead uses the accessible binary companion. The two general statements share the odd-field argument and should be credited and reviewed with that shared dependency visible.

No new finite experiment or end-to-end Lean verification is offered by this reduction note. Compilation and package hashes, when supplied, associate the document with its source; they do not certify the asymptotic theorem. Full Lean formalization remains unfinished. The finite-field quantifiers, ordinary coefficient representation and tape costs are part of any future statement-level verification.

References

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